

# MAX ZHU

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## EDUCATION

### Georgia Institute of Technology

B.S. in Computer Engineering, Minor in Robotics

- Threads: Computing Hardware & Emerging Architecture and Systems & Architecture

Atlanta, GA

Aug 2023 – Present

GPA: 3.9/4.0

## TECHNICAL SKILLS

**Languages:** C++, C, Python, Java, Perl, TCL, JavaScript, MATLAB, HTML, SQL

**Frameworks/Tools:** Django, React, Git, RISC-V, SPICE, GTKWave, Docker, Heroku

**Hardware/Systems:** SystemVerilog, UVM, VHDL, FPGA, RTL, PCB Design, ARM, CAD, Arduino, Oscilloscopes

## EXPERIENCE

### FUSMobile

Embedded and QA Engineer

Alpharetta, GA

Apr 2025 – Present

- Built an automated medical report generation system in **C++**, using **MFC** for metadata extraction, **OpenCV** for image filtering/processing, and **hpdf** for annotation, eliminating manual reporting time
- Developed robust **serial communication protocol** between **Arduino Due** and imaging workstation, allowing for plug and play capability and hands-free self-alignment
- Re-engineered the Neurolyzer suite prioritizing portability and interactivity, for hands-on conference demonstrations

### CRNCH Rogues Gallery

Reconfigurable Researcher

Atlanta, GA

January 2025 – May 2025

- Created and containerized the **XChaCha20** encryption algorithm with **Verilog** and **Docker**
- Offloaded cryptographic operations from the CPU to a **PYNQ FPGA**, achieving a 10x efficiency improvement

### MARGA Lab

Wireless Digital Twin Researcher

Atlanta, GA

Aug 2024 – May 2025

- Created scripts in **Python** using **Sionna** to simulate and optimize transmitter-receiver placement by over 60%
- Analyzed and annotated 50+ research papers on **5G**, **Wi-Fi 6**, **ray tracing**, **wireless signals**, and **UAVs**
- Automated geometric simplification of Blender environments, reducing simulation rendering time by 50% while maintaining over 95% signal fidelity compared to full-scale models

## PROJECTS

### RISC-V Pipelined Processor | SystemVerilog, GTKWave, RISC-V

- Created a 5-stage **RISC-V** pipelined processor, complete with forwarding and hazard detection in **SystemVerilog**
- Debugged waveforms with **GTKWave** and handled structural, data, and control hazards

### SCOMP Processor & Memory Device | VHDL, DE-10 Standard, SCOMP

- Created and documented a SCOMP processor and memory device with **VHDL** on a **DE-10** FPGA board

### Spotify Car Thing + Wrapped Web App | ESP8266, Django, OAuth, HTML/CSS/JS

- Built a DIY Spotify Car Thing IoT dashboard, displaying, skipping, and liking songs with an **ESP8266** and **C++**
- Designed a personalized analytics web app with **Django**, Spotify API, OAuth, and CSS animations

## EXTRACURRICULARS

### GT Institute of Electrical Engineers | Director of Corporate Affairs

May 2025 – Present

- Facilitated general body meetings, networking sessions, and career talks with 100+ students
- Secured industry sponsorships and alumni speakers to expand engagement opportunities

### Silicon Jackets | Verification Engineer & Digital Designer

Aug 2024 – Present

- Designed and verified custom **RISC-V** ALU, load-store unit, and data-cache pipeline in **SystemVerilog**
- Developed and synthesized 500+ automated tests in **Python** to verify core RTL functionality

### HIVE Makerspace | Master Peer Instructor

Dec 2023 – Present

- Led hands-on workshops in laser cutting, PCB fabrication, 3D printing, and woodworking
- Maintained and repaired equipment including sewing machines and PCB fabrication tools